Smart Independent Living Assistant: A Low-Cost Integrated IoT System for Elderly Safety and Independent Living

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***Abstract*—Elderly people who live alone face a combination of everyday risks—missed medication, kitchen gas and fire accidents, unauthorized entry, and falls with no one nearby to help—while caregivers cannot monitor them continuously. Existing assistive products typically address only one or two of these risks, and most medicine systems stop at simple reminders. This paper presents the *Smart Independent Living Assistant*, a low-cost IoT prototype that integrates four safety functions in one smart-home unit on an Arduino UNO, an ESP32, and an ESP32-CAM. The system follows a *Sense* $\rightarrow$ *Decide* $\rightarrow$ *Act* $\rightarrow$ *Notify* pipeline: it unlocks the correct medicine drawer at the scheduled time and raises a missed-dose alert if the drawer is not opened within a 5–10 minute window; it detects gas, smoke, or flame and automatically closes the gas valve, starts an exhaust fan, and sounds an alarm; it authenticates visitors by RFID and captures an image when repeated unauthorized attempts occur; and it detects falls through an inertial sensor or a manual SOS button and alerts the caregiver over Wi-Fi. Section IV specifies the functional scenarios and evaluation protocol. The accompanying diagrams illustrate the intended responses; recorded trials and notification timestamps were not available for this report, so detection rates and alert latency are not claimed.**

***Index Terms*—Internet of Things, elderly care, smart home, medication adherence, fall detection, gas leakage detection, ESP32, Arduino**

I. Introduction

A. Problem Statement

Population ageing is increasing the number of older adults who wish to remain in their own homes rather than move into care facilities. Living independently, however, exposes them to a cluster of everyday hazards. First, **medication adherence** is difficult: the World Health Organization reports that adherence to long-term therapy in chronic illness averages only about 50% in developed countries [7], and forgetting a dose or taking it at the wrong time is a common cause. Second, **kitchen accidents**—unattended stoves, gas leaks, smoke, and fire—are a particular danger for people with failing memory or reduced sense of smell. Third, **home security** is a concern, since older adults living alone are vulnerable to unauthorized entry. Fourth, **falls** are the second leading cause of unintentional injury deaths worldwide, and adults older than 60 years suffer the greatest number of fatal falls [6]; the consequences are far worse when nobody is nearby to help. Finally, **caregivers** such as family members cannot be physically present or watch continuously.

These observations lead to the central question of this project: *how can IoT technology provide continuous safety monitoring and automatic assistance for elderly people, while reducing their dependence on caregivers?* We chose this problem because it combines concrete, testable embedded-systems tasks (sensing, real-time decision logic, actuation, wireless communication) with a real social need, and because the required technology—microcontrollers and inexpensive sensors—is affordable enough to be deployed in ordinary households.

B. Literature Review

A substantial body of work addresses individual aspects of elderly assistance, mostly on low-cost microcontroller platforms. We group the prior work into four themes.

Integrated embedded assistants. Bernardo [1] proposed an embedded system for monitoring the elderly that combines fall detection (accelerometer and barometer), gas-leak detection at home, and a medicine dispenser with separate compartments, adding message sending and IP-camera video transmission. This is the closest precedent to our work, but it is built around a comparatively expensive embedded board and does not include access control or a graded intruder response.

Medicine reminders and environment monitoring. A recent ESP32-based system couples a medicine reminder with kitchen environmental monitoring (temperature, humidity, gas, air quality) and a mobile interface [2]. Its authors note that

current elderly-care solutions are often fragmented, costly, or poorly customizable—the gap that motivates integration. Such systems, however, generally issue reminders only; they do not physically control access to the medicine or react to a missed dose.

Fall detection. Many works pair an MPU-6050 inertial sensor with an ESP32. Farhan *et al.* [3] add a GPS module and send a position link and caregiver alert when a fall occurs. A wearable ESP32/MPU-6050 system with a Node-RED dashboard [4] found that fixed acceleration thresholds alone could not reliably separate fall classes from daily movements, and also observed supply instability when powering the ESP32 from a single battery cell. A mobile IoT monitoring system [5] combines fall detection, indoor positioning, and appliance control. These works confirm the feasibility of low-cost fall detection but stress the need for threshold tuning, a manual fallback, and a robust power supply.

Smart-home security. RFID-based door locks and PIR-based presence detection are widely used in hobbyist and academic prototypes, but are rarely integrated with the health-and-safety functions above.

Table I summarizes the coverage of the reviewed works compared with the proposed system. A dash means that the function is not reported in the material we reviewed.

TABLE I
FUNCTIONAL COVERAGE OF RELATED WORK VERSUS THE PROPOSED SYSTEM

Work	Medicine	Gas/Fire	Door	Fall/SOS
Bernardo [1]	✓	✓	–	✓
UTHM system [2]	✓	✓	–	–
Farhan *et al.* [3]	–	–	–	✓
Wearable ESP32 [4]	–	–	–	✓
Mobile IoT [5]	–	–	–	✓
Proposed	✓	✓	✓	✓

C. Innovative Features of the Proposed System

Compared with existing solutions, our system adds the following features.

1) **Compliance-aware medicine management.** Instead of a reminder only, an RTC and reed switch track time and drawer state; the correct drawer is unlocked at dose time, a 5–10 minute timer starts, and if the drawer is not opened the drawer is re-locked and a *missed-dose alert* is sent to the caregiver.
2) **Automatic kitchen hazard response.** Detection of gas, smoke, or flame does not merely sound an alarm: a servo closes the gas valve, a fan is switched on, and a buzzer, red LED, and OLED warning are activated.
3) **Graded smart door security.** Authorized RFID cards grant entry, while repeated unauthorized attempts trigger an intruder mode with an alarm, a camera snapshot from the ESP32-CAM, and a caregiver alert.
4) **Automatic plus manual emergency detection.** A fall is detected from inertial data without user action, with a manual SOS button as a fallback for events the sensor cannot detect.
5) **One integrated, low-cost unit** built from open-source Arduino/ESP32 platforms, with local fallback logic so that core safety functions keep working if Wi-Fi is lost.

II. PROPOSED METHOD

A. Block Diagram of the Overall System

Fig. 1 shows the overall architecture. The design follows a *Sense* $\rightarrow$ *Decide* $\rightarrow$ *Act* $\rightarrow$ *Notify* pipeline. Sensors for the four safety functions (left) feed the Arduino UNO, which serves as the central real-time controller. The Arduino drives the local actuators and indicators (right) and passes event messages to the ESP32 over a serial link. The ESP32 provides Wi-Fi connectivity, triggers the ESP32-CAM when an image is needed, and delivers notifications to the caregiver.

B. Description of the Blocks

1) Sensing blocks: The **medicine module** uses a DS3231 real-time clock (RTC) as the time base. The DS3231 is a temperature-compensated RTC with an I^2C interface [12], so schedule times remain accurate without Wi-Fi. A reed switch on each drawer reports whether it is open or closed.

The **kitchen module** uses an MQ-2 sensor and a flame sensor. The MQ-2 is a metal-oxide semiconductor sensor that responds to LPG, propane, hydrogen, methane, and smoke through a change in sensing resistance [9]; the analog output is sampled by the ADC of the Arduino and compared with a calibrated threshold. The IR flame sensor provides an independent indication of open flame.

The **door module** combines a PIR sensor (presence near the entrance), an RC522 RFID reader (identity), a reed switch (door open/closed), and the ESP32-CAM (visual evidence). The RC522 is based on the NXP MFRC522 and supports ISO/IEC 14443 A contactless cards at 13.56 MHz over SPI [11].

The **fall/emergency module** uses an MPU6050 six-axis inertial sensor (3-axis accelerometer and 3-axis gyroscope) [10] to detect abnormal movement, and a push-button SOS for manual activation.

2) Control blocks: The **Arduino UNO** (ATmega328P) is the central controller. It polls all sensors, runs the timers and decision logic for the four features, and drives the actuators. Placing time-critical safety decisions on the Arduino means that the gas valve, door lock, and alarms do not depend on network availability.

The **ESP32** [8] is the connectivity block. It receives short event messages from the Arduino (e.g., `MISSED_DOSE`, `GAS_ALERT`, `INTRUDER`, `FALL`, `SOS`) and forwards them to the caregiver over Wi-Fi. The **ESP32-CAM** captures an image when the ESP32 signals an intruder event, and the image is attached to the caregiver alert. Because Arduino UNO logic levels are 5 V while the ESP32 is 3.3 V, a level shifter or voltage divider is used on the serial link from the Arduino to the ESP32.

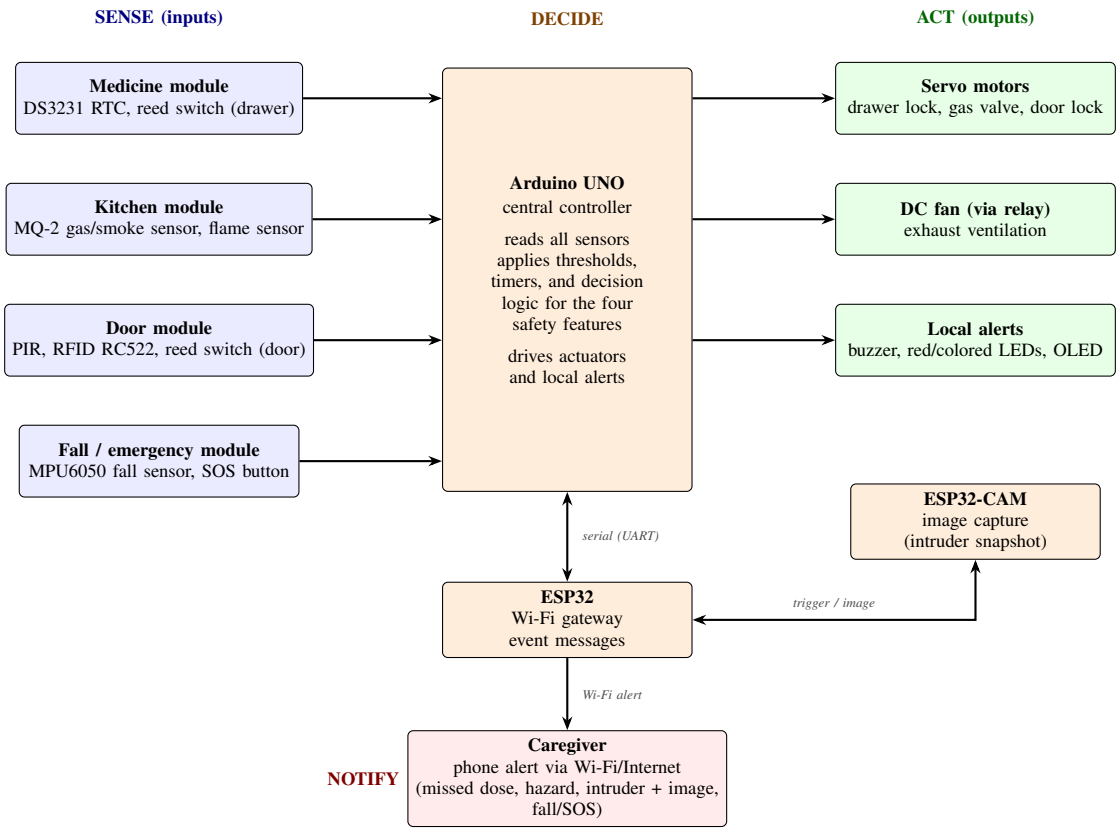


Fig. 1. Block diagram of the Smart Independent Living Assistant. Sensors report to the Arduino UNO, which decides and drives local actuators; the ESP32 relays events to the caregiver and triggers the ESP32-CAM for intruder snapshots.

3) Output blocks: **Servo motors** actuate three mechanisms: the medicine drawer latch, the gas valve, and the door lock. A **DC fan** driven through a **relay module** provides exhaust ventilation; the relay isolates the fan's current and supply from the controller. The **OLED display**, **buzzer**, and **red/colored LEDs** give the elderly user immediate local feedback (reminders, warnings, and the word "EMERGENCY").

C. Operating Logic of the Four Safety Features

Each feature follows the same Information → Decision → Action pattern, summarized in Table II. The corresponding algorithms are given in Algorithms 1–4.

The numeric thresholds (θ_g, θ_{ff}, θ_{imp}, T_i, N, K, W) are configuration constants in the firmware. They are set initially from datasheet guidance and then tuned experimentally, which directly addresses the false-alarm and missed-detection challenge identified in the project plan. The multi-sample confirmation (N) and the inactivity check (T_i) are the first line of defence against false alarms, while the SOS button covers missed detections.

III. Implemented Hardware System

A. List of Hardware and Software Environment

Table III lists the hardware used. The components are inexpensive, widely available, and supported by open-source libraries, which keeps the overall cost low.

Algorithm 1 Smart medicine compliance

Require: schedule $\{(t_i, d_i)\}$ of dose times and drawers; window $W \in [5, 10]$ min

loop
  $t \leftarrow$ RTC.now()
  if $t = t_i$ for some i **then**
    unlock drawer d_i (servo); show reminder on OLED; sound buzzer
    start timer $T \leftarrow W$; *taken* $\leftarrow$ false
    while $T > 0$ **and not** *taken* **do**
      if reed switch of d_i reports OPEN **then**
        taken $\leftarrow$ true; stop buzzer; log dose as taken
      end if
    end while
    re-lock drawer d_i once it is closed (or when T expires)
    if not *taken* **then**
      log missed dose; send `MISSED_DOSE` to ESP32
    end if
  end if
end loop

Software environment. Firmware for the Arduino UNO and ESP32 boards was written in Embedded C/C++ using the **Arduino IDE**. The ESP32 uses its built-in **Wi-Fi** stack for communication, and the ESP32-CAM uses the camera driver provided in the ESP32 Arduino core. Standard open-source libraries are used for the peripherals (`Wire`, `SPI`, `Servo`, an RTC library for the DS3231, an MFRC522 library, an MPU6050 library, and Adafruit GFX/SSD1306 for the OLED). The ESP32 is responsible for forwarding caregiver

TABLE II
INFORMATION–DECISION–ACTION SUMMARY OF THE FOUR SAFETY FEATURES

Feature	Information (sense)	Decision	Action and notification
1. Medicine compliance	DS3231 RTC time; drawer reed switch	Medicine time reached → unlock the correct drawer and start a 5–10 min timer	Servo unlocks drawer; OLED and buzzer remind the user. If not opened in time: re-lock drawer and send missed-dose alert
2. Kitchen safety	MQ-2 gas/smoke and flame sensor readings	Reading above safe threshold, or flame detected	Servo closes gas valve; fan ON; buzzer and red LED activate; OLED shows warning; caregiver alert
3. Door security	PIR, RFID RC522, door reed switch, camera	Authorized RFID → allow entry; unauthorized or repeated attempts → intruder mode	Servo controls lock; alarm sounds; ESP32-CAM captures the intruder; ESP32 alerts caregiver
4. Fall and emergency	MPU6050 motion data; SOS button	Fall detected or SOS pressed → emergency mode	Buzzer sounds; red LED flashes; OLED shows EMERGENCY; ESP32 notifies caregiver

TABLE III
HARDWARE COMPONENTS USED IN THE PROTOTYPE

Category	Component (model)	Role in the system
Controller	Arduino UNO R3 (ATmega328P)	Central real-time controller: sensor polling, timers, decision logic, actuator control
Controller	ESP32 development board (ESP32-WROOM-32)	Wi-Fi gateway; formats and sends caregiver notifications
Controller	ESP32-CAM (AI-Thinker, OV2640 camera)	Captures an image of the intruder on request
Sensor	DS3231 RTC module	Accurate time base for medicine schedule (I^2C)
Sensor	MQ-2 gas/smoke sensor module	Detects LPG/gas and smoke (analog output)
Sensor	IR flame sensor module	Detects open flame (digital/analog output)
Sensor	PIR motion sensor (HC-SR501)	Detects a person near the entrance
Sensor	RFID reader RC522 (MFRC522, 13.56 MHz) with cards/tags	Identifies authorized users (SPI)
Sensor	Magnetic reed switches	Drawer and door open/closed status
Sensor	MPU6050 (GY-521) six-axis IMU	Fall detection from acceleration data (I^2C)
Input	Push-button	Manual SOS
Actuator	Micro servo motors (SG90/MG90S class)	Drawer latch, gas valve, and door lock
Actuator	5 V DC fan with 5 V relay module	Exhaust ventilation during gas events
Output	$0.96''$ OLED display (SSD1306, 128×64, I^2C)	Reminders, warnings, and emergency messages
Output	Active buzzer; red/colored LEDs	Audible and visual alarms
Power	5 V regulated supply (separate rail for servos/fan)	Powers motors without disturbing logic circuits

Algorithm 2 Kitchen hazard response

Require: gas threshold θ_g; confirmation count N
loop
  $g \leftarrow$ ADC(MQ-2); $f \leftarrow$ flame sensor state
  if ($g > \theta_g$ for N consecutive samples) **or** (f = DETECTED for N samples) **then**
    close gas valve (servo); fan ON (relay)
    buzzer ON; red LED ON; OLED: "GAS/FIRE WARNING"
    send `GAS_ALERT` to ESP32
  else if $g \leq \theta_g - \delta$ and no flame **then**
    alarms OFF; fan OFF after clearing delay
  end if
end loop

Algorithm 3 Smart door security

Require: authorized UID list; attempt limit K; counter $c \leftarrow 0$
loop
  if PIR detects presence **and** card presented **then**
    if UID $\in$ authorized list **then**
      unlock door (servo); log entry; $c \leftarrow 0$; re-lock after delay
    else
      $c \leftarrow c + 1$; short buzzer warning
    end if
  end if
  if $c \geq K$ **or** door reed switch OPEN while locked **then**
    intruder mode: alarm ON; ESP32-CAM captures image
    send `INTRUDER` + image to caregiver via ESP32
  end if
end loop

notifications over Wi-Fi. The project materials do not identify the application or messaging service used, so delivery-channel implementation and end-to-end receipt remain unverified in this report.

B. Interfacing and Interdependencies

Table IV shows how the peripherals are connected. Placing the I^2C devices on a shared bus reduces wiring; the MPU6050 address is moved to 0x69 (AD0 pin high) so that it does not

Algorithm 4 Fall and emergency detection

Require: free-fall threshold θ_{ff}; impact threshold θ_{imp}; inactivity time T_i

loop
  $a \leftarrow \sqrt{a_x^2 + a_y^2 + a_z^2}$ (MPU6050, in g)
  if $a < \theta_{ff}$ followed within Δt by $a > \theta_{imp}$ **then**
    if little motion for T_i seconds **then**
      emergency $\leftarrow$ true
    end if
  end if
  if SOS button pressed **then**
    emergency $\leftarrow$ true
  end if
  if *emergency* **then**
    buzzer ON; red LED flashing; OLED: "EMERGENCY"
    send `FALL` or `SOS` to ESP32 for caregiver alert
  end if
end loop

collide with the DS3231, which also uses 0x68.

TABLE IV
INTERFACE ORGANIZATION OF THE PROTOTYPE

Interface	Devices	Notes
I^2C	DS3231, MPU6050, SSD1306 OLED	Addresses 0x68, 0x69, 0x3C [13]
SPI	RC522 RFID	3.3 V device
Analog in	MQ-2	Sampled by ADC
Digital in	Flame, PIR, reed switches, SOS	Debounced in software
PWM out	Three servos	Powered from separate 5 V rail
Digital out	Relay (fan), buzzer, LEDs	Relay isolates the fan load
UART	Arduino ↔ ESP32	Level shifting 5 V→3.3 V

C. Figures of the Implemented Hardware System

Fig. 2 shows the functional layout of the prototype, Fig. 3 gives the wiring diagram with pin assignments, and Fig. 4 provides component-level illustrations of the four feature modules. These drawings describe the design and are not photographs of an assembled prototype.

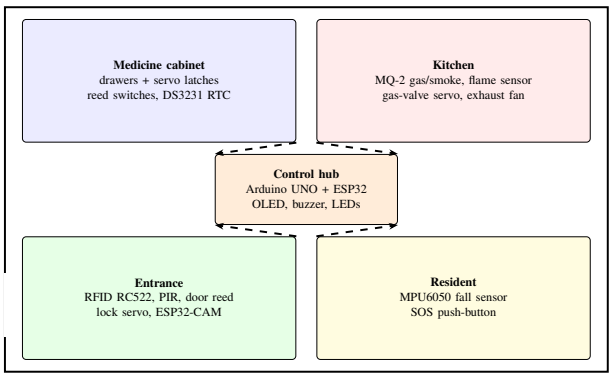


Fig. 2. Functional layout of the prototype showing where the four safety modules are placed and how they connect to the control hub.

IV. Functional Scenarios and Evaluation Plan

This section describes the intended functional responses and the protocol for evaluating each module and the complete alert path. No measurement logs, trial counts, prototype photographs, or caregiver screenshots were supplied with the project files. Consequently, the figures in this section are explanatory diagrams, not experimental evidence, and Table V distinguishes the expected outcomes from unavailable measurements.

A. Test Method

Each feature should be evaluated in two phases: a *normal* condition in which no event should be raised (to check for false alarms) and an *event* condition in which the specified response should follow (to check for missed detections). The serial log should record the sensor readings, decision state, actuator state, and event timestamp; caregiver receipt should be recorded separately. For each scenario, report the trial count and pass count, with success rate equal to $100\, n_{\text{pass}}/n_{\text{trials}}$. Alert latency is the elapsed time from event detection to confirmed caregiver receipt, using synchronized timestamps. Repeat the local-response checks with Wi-Fi disconnected and distinguish local success from notification failure.

B. Medicine Compliance

With the RTC set to a dose time, the correct drawer should unlock, the OLED and buzzer should remind the user, and the timer should start. The two evaluation cases are: (i) the drawer is opened within the window, so the opening is logged and the reminder stops; (ii) the drawer is not opened, so it re-locks and a missed-dose event is sent to the ESP32. Fig. 5 illustrates both paths. A reed switch confirms access to the drawer only; it cannot establish that the prescribed dose was ingested.

C. Kitchen Safety

The intended normal state has an MQ-2 reading below the configured threshold and no active flame indication. When either input satisfies the confirmation rule, the controller commands the valve servo to close, activates the fan relay, and displays a warning with the buzzer and red LED. Fig. 6 illustrates the threshold and response sequence without measured data. Bench verification should begin with injected sensor signals and a disconnected demonstration valve/fan, rather than releasing combustible gas or creating an open flame. The illustrated response is prototype logic, not a validated procedure for a real gas emergency.

D. Door Security

The intended authorized path accepts a registered card and commands the lock servo to open; an unregistered card is rejected and counted. Reaching the attempt limit K, or detecting an open door while locked, should activate intruder mode, sound the alarm, request an ESP32-CAM image, and forward an alert to the caregiver. Fig. 7 separates these paths. Evaluation must check image capture and caregiver receipt independently of the local lock/alarm response.

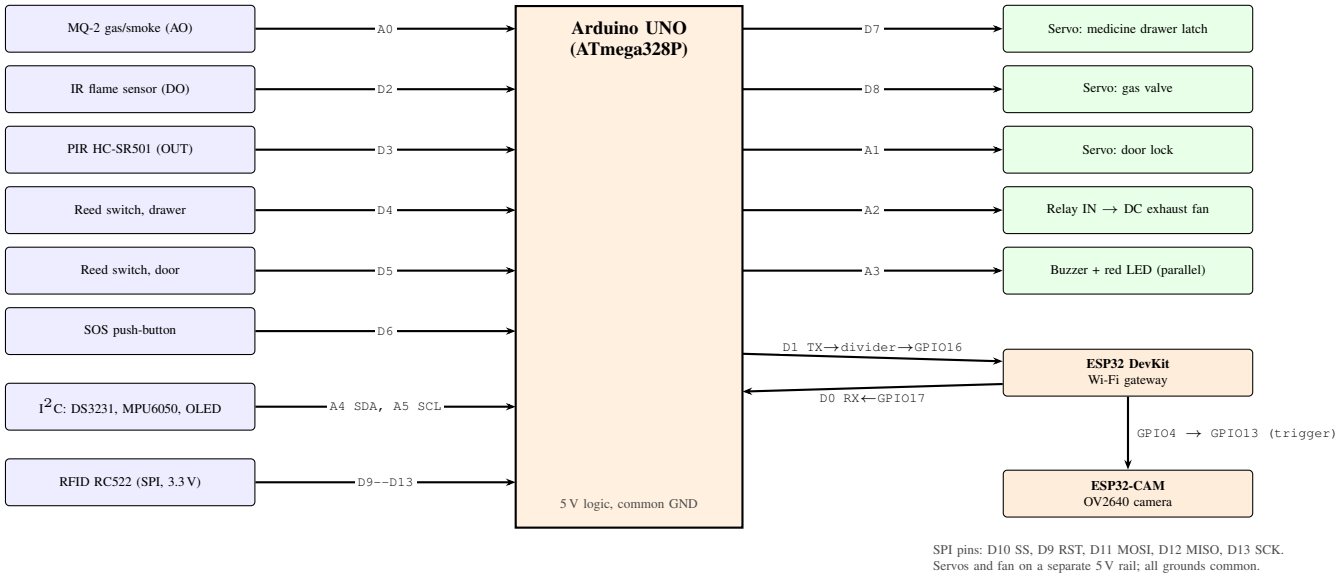


Fig. 3. Wiring diagram of the prototype with Arduino UNO pin assignments. Sensors enter on the left; actuators and the ESP32 link are on the right.

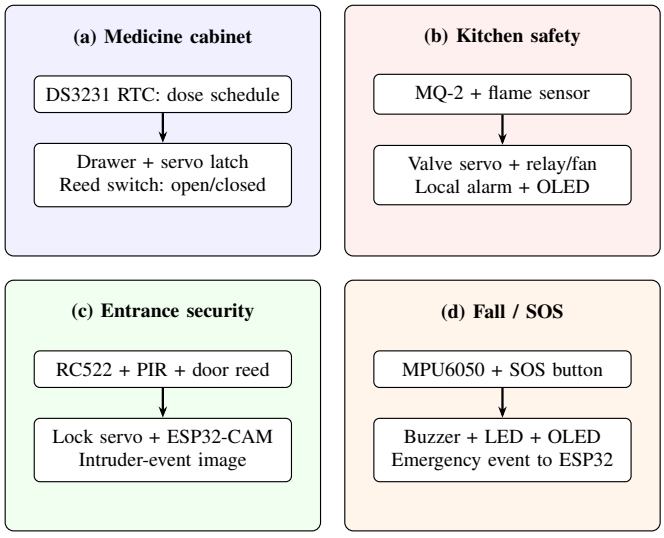


Fig. 4. Component-level design illustrations: (a) medicine drawer, (b) kitchen safety, (c) door security, and (d) fall/SOS. These are schematic drawings, not prototype photographs.

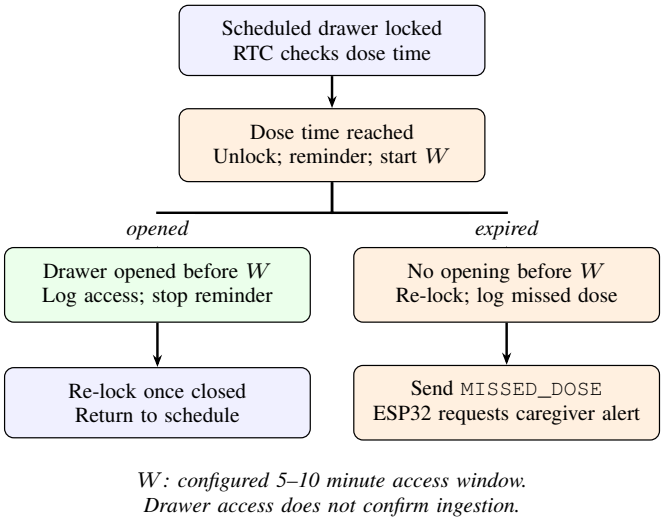


Fig. 5. Intended medicine-management state transitions. Drawer opening is an access event, not proof of medication ingestion; no measured trial is depicted.

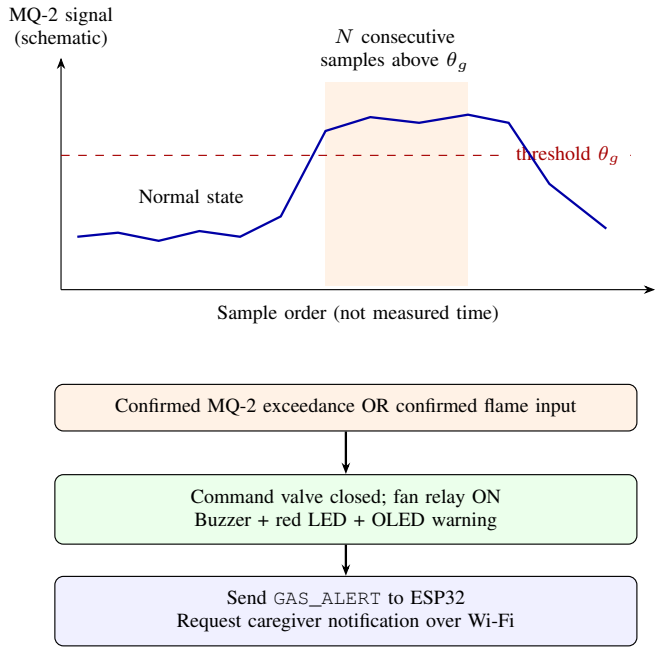
E. Fall and Emergency Detection

The detector looks for a low-acceleration interval followed by an impact peak and then low motion; the conceptual pattern in Fig. 8 explains the algorithm rather than demonstrating detection accuracy. Bench evaluation should use a safely handled sensor or padded test fixture, not deliberate falls by an older adult, and should include normal movements to measure false alarms. The SOS button bypasses the inertial decision path. Either event should activate the buzzer, flashing red LED, OLED emergency message, and ESP32 notification request; caregiver receipt requires separate verification.

F. Evaluation Coverage and Evidence Status

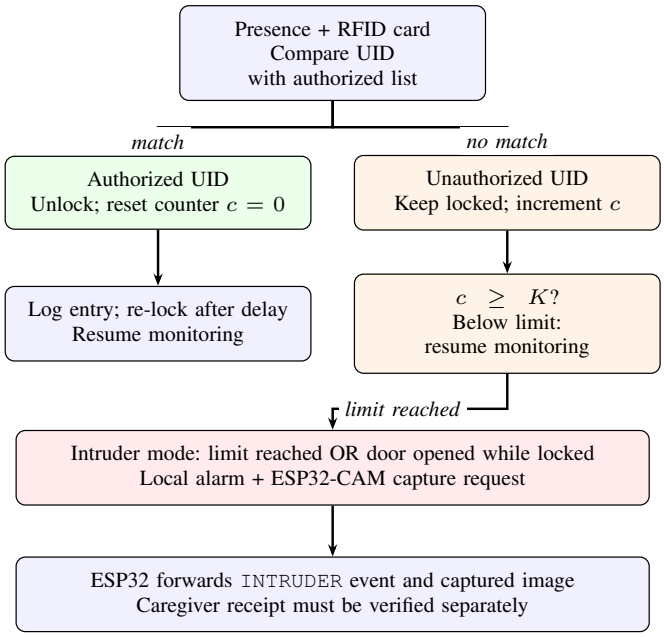
Table V lists the required checks and the status of the supporting evidence. NR means not reported: it is neither a zero trial count nor a failed test. Local actuator checks, notification delivery, and quantitative latency should be recorded separately.

No numerical success rate, false-alarm rate, or alert latency can be calculated from the supplied files. A completed evaluation should report passes and failures for each scenario, false alerts during normal activity, and the distribution of delivery latency in seconds (including failed deliveries). The diagrams



Prototype response sequence; not validated gas-safety equipment.

Fig. 6. Conceptual kitchen hazard detection and response. The trace is schematic, without measured ADC values or timing; thresholds and confirmation count require calibration.



K: configured unauthorized-attempt limit.

Fig. 7. Intended door-security decision paths and intruder notification sequence. This diagram is not an access log, camera image, or proof of alert delivery.

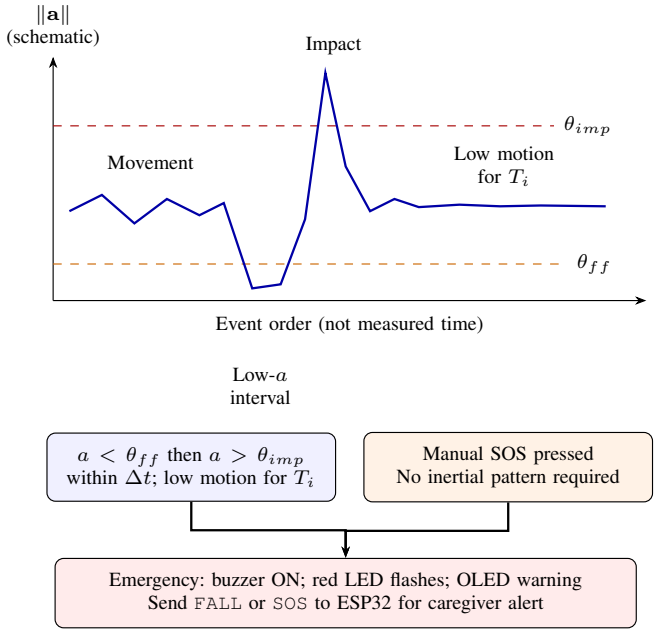


Fig. 8. Conceptual fall-detection pattern and manual SOS path. The acceleration trace is illustrative rather than a recorded fall, and neither amplitudes nor durations are measured.

TABLE V
FUNCTIONAL EVALUATION CHECKLIST; NR = NOT REPORTED

Test	Expected outcome	Trials	Passes
Dose reminder + unlock	Drawer unlocks at dose time	NR	NR
Missed-dose alert	Re-lock and alert after window	NR	NR
Gas/smoke response	Valve closes, fan on, alarm	NR	NR
Flame response	Same as gas response	NR	NR
Authorized entry	Door unlocks	NR	NR
Intruder mode	Alarm, image, alert	NR	NR
Fall detection	Emergency mode	NR	NR
SOS button	Emergency mode	NR	NR
Alert delivery	Caregiver receipt verified	NR	NR

establish the intended behavior only; they do not validate reliability or clinical effectiveness.

V. DISCUSSION

A. Limitations

The prototype has several limitations. (1) The MQ-2 is non-selective, so cooking vapors, alcohol, or aerosols can raise the reading and trigger a false gas alarm, and the sensor needs a warm-up period and calibration before its readings are reliable [9]. (2) The IR flame sensor can be disturbed by strong sunlight or other infrared sources, which may cause false flame detection. (3) Threshold-based fall detection can misclassify abrupt sitting as a fall and may miss slow, gradual falls; similar limits have been reported for fixed thresholds in the literature [4]. The motion sensor also works only when the person carries it, and the SOS button only helps if it can be reached.

(4) RFID identification based on card UID is convenient but a card can be lost or copied, so it is weaker than multi-factor authentication. (5) Notifications depend on Wi-Fi; when the connection fails, only local alarms and actuation remain. (6) The servo-based gas valve is a prototype mechanism and would need a certified shut-off valve in a real installation.

B. Conflicting Requirements and Trade-offs (P2)

The project has several competing requirements. *Low cost versus reliability:* inexpensive sensors such as the MQ-2 and IR flame sensor are less selective than industrial detectors, so reliability was recovered through software (multi-sample confirmation and tuned thresholds) rather than costlier hardware. *Fast response versus false-alarm avoidance:* requiring several consecutive samples before acting delays the alarm slightly but avoids spurious valve closures. *Security versus privacy:* a camera improves intruder evidence but can intrude on the residents' privacy, so the camera is used only to capture a snapshot on an intruder event rather than streaming continuously.

C. Alternative Designs (P3)

Alternative 1: a Raspberry Pi hub with machine-learning analysis. A single-board computer could run a trained fall-detection model and face recognition on camera images, giving higher accuracy and richer logs. It would, however, cost more, consume more power, require an operating system with slower boot and less deterministic timing, and be harder for non-technical caregivers to maintain. *Alternative 2: a single ESP32 doing everything.* A single ESP32 would reduce the board count and cost, but the available GPIO pins and the 3.3 V logic would constrain the number of sensors and servos, and a Wi-Fi fault or software stall could affect the time-critical safety logic. Our three-board design costs slightly more than the single-board option but keeps safety decisions on the Arduino independent of the network, and is cheaper and simpler than the Raspberry Pi approach.

D. Impact and Familiarity of Issues (P4)

Although the implementation is embedded systems work familiar to a computer-engineering student, the problem domain is not: it required learning about elderly care, medication adherence, fall risk, and kitchen and home safety, and designing for users who may be uncomfortable with technology. This is why the interface is deliberately simple (OLED text, buzzer tones, colored LEDs) and why caregivers only receive alerts when action is needed. If deployed, the system could reduce the daily risk to older adults, delay the need for institutional care, and relieve some burden on family caregivers.

E. Standards and Ethics (P5)

The design relies on published interface standards and manufacturer documentation: the I^2C bus specification [13], SPI, UART, IEEE 802.11 Wi-Fi on the ESP32 [8], ISO/IEC 14443 A on the RC522 [11], and the datasheets of the DS3231, MPU6050, and MQ-2. The prototype deviated from the documentation in one practical respect: the 5 V Arduino and 3.3 V ESP32/RC522 devices required level shifting. Ethically, the system collects sensitive information (health routines, movement, and images), so the data collected is limited to what is needed, the camera records only on intruder events, and the elderly person's consent and the caregiver's role should be agreed in advance. The prototype is an assistive aid and is not a certified medical or life-safety device; it should supplement, not replace, human care and certified gas/smoke alarms.

F. Stakeholders (P6)

Two main stakeholders were identified: the elderly resident, who needs a non-intrusive and easy-to-understand device, and the caregiver, who needs timely and trustworthy alerts without constant monitoring. These needs shaped the design: local feedback through the OLED and buzzer, notification only on defined events, and a manual SOS option.

G. Interdependence of Modules (P7)

The system is modular: each safety feature has its own sensors and actuators and its own logic block in the firmware, so one feature can be tested or modified without affecting the others. The modules are, however, coupled through shared resources: all features depend on the Arduino and share the I^2C bus, the power rails, and the buzzer/OLED. Hardware and software on separate devices also depend on each other: the Arduino produces events that the ESP32 turns into notifications, and the ESP32-CAM image depends on a trigger from the ESP32. This coupling is limited to a simple serial message protocol, and the Arduino continues its local safety actions even if the ESP32 is unavailable.

P1	P2	P3	P4	P5	P6	P7
✓	✓	✓	✓	✓	✓	✓

TABLE VI
COMPLEX ENGINEERING PROBLEM MAPPING. P1 – DEPTH OF KNOWLEDGE, P2 – CONFLICTING REQUIREMENTS, P3 – DEPTH OF ANALYSIS, P4 – FAMILIARITY OF ISSUES, P5 – EXTENT OF APPLICABLE CODES, P6 – EXTENT OF STAKEHOLDERS, P7 – INTERDEPENDENCE.

VI. CONCLUSION

This paper presented the Smart Independent Living Assistant, a low-cost integrated IoT prototype that combines medicine compliance, kitchen hazard response, door security, and fall/emergency detection in one Arduino–ESP32 system, with automatic local action and caregiver notification. By enforcing a dose window with a missed-dose alert and by pairing automatic fall detection with a manual SOS, it goes beyond the reminder-only or single-function systems reviewed in the literature. The evaluation checklist covers all four functions and the caregiver delivery path, but recorded trials and receipt timestamps are not available. Accordingly, this report does not claim a measured detection success rate, response time, or reduction in caregiver burden; these remain objectives for experimental validation.

Future work will focus on field-testing the prototype in real households, tuning sensor thresholds to further reduce false

alarms, adding battery backup for the core functions, replacing fixed fall thresholds with a learned classifier, developing a caregiver-friendly mobile application and dashboard, and strengthening door authentication with a secondary factor such as a PIN.

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